Assignment-Regression Algorithm

Problem Statement or Requirement:

As a Data Scientist, I could identify certain solutions for the problem:

1. Problem Statement – Predict the insurance Charges
2. Problem Identification:
   Stage 1: Machine Learning
   Stage2: Supervised Learning
   Stage3: Regression
3. Dataset: 1. Total no of rows:1338
   2. Total no of columns: 6
   3. Output is charges (Last column)
4. Pre Processing Method: This is a Multi Linear Regression / Sex & the smoker to be changed to Categorized Data as Python takes only 1 & 0.
5. Multi Linear Regression: 0.7894
6. Support Vector Machine
   (keeping epsilon =0.1)

S.No	Hyper Parameter	Linear (r Value)	RBF(Non-Linear)- r value	Poly (r Value)	Sigmoid (r value)
1	C10	0.42	-0.03	0.038	0.393
2	C100	0.62	0.32	0.617	0.527
3	C500	0.763	0.664	0.826	0.444
4	C1000	0.764	0.81	0.856	0.287
5	C2000	0.744	0.854	0.860	-0.59
6	C3000	0.741	0.866	0.859	-2.12

Support Vector Machine = r value = 0.866 (when Kernel = rbf, C=3000, Epsilon =0.1)

7. Decision Tree: 0.755
   (data is attached separately)

8. Random Forest: 0.8498

   Conclusion:
   According to me, SVM method gave a better result & I would like to deploy this method for better r value.